CYBER SECURITY INTERNSHIP

WEEK 5 TASK

Vulnerability Assessment & Penetration Testing (VAPT)

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Organization	DG Interns Hub
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Environment	Kali Linux lab; authorized training target

Practical scope: Nmap reconnaissance, Nikto web scanning, and Burp Suite HTTP request inspection. Testing was limited to the designated training target and local proxy workflow.

Prepared as a practical learning report based on the terminal and application screenshots captured during the session.

1. Executive Summary

This Week 5 practical introduced a basic VAPT workflow using Nmap, Nikto, and Burp Suite Community Edition. Nmap completed three scan modes against the assigned training hostname. The scans reported the host as up, but all 1,000 default TCP ports were filtered/no-response in the observed results. No open ports or service versions were confirmed by those scans.

Nikto could not establish a connection to the target on TCP/80. Burp Suite's local proxy listener was configured on 127.0.0.1:8080 and HTTP history showed GET requests for the target URL, but the browser displayed a connection error and no successful target response was observed. Network checks showed the NAT gateway and 1.1.1.1 were pingable; DNS and web connectivity were inconsistent.

Key outcome

The practical successfully demonstrated scan execution, proxy setup, request inspection, and troubleshooting evidence. It did not confirm exploitable vulnerabilities; findings below distinguish observations from unverified possibilities.

Scope and safety

Target: testphp.vulnweb.com (training site named in the assignment). Activities were limited to basic reconnaissance and HTTP request inspection. No exploitation, credential attacks, or intrusive actions were performed.

Evidence summary

Area	Observed result	Status
Nmap -sS	Host up; 1,000 TCP ports filtered	Completed
Nmap -sV	Host up; 1,000 TCP ports filtered; no versions	Completed
Nmap -A	Host up; no exact OS match; all 1,000 TCP ports filtered	Completed
Nikto	Unable to connect to target TCP/80	Blocked by connectivity
Burp Suite	Proxy listener active; GET requests visible in history	Partially completed

2. Introduction and Methodology

Vulnerability Assessment and Penetration Testing (VAPT) is a structured way to identify and validate security weaknesses. Vulnerability assessment looks for possible issues; penetration testing validates impact under authorized conditions. This practical focused on introductory reconnaissance and request analysis rather than exploitation.

Tools used

Tool	Purpose in this practical
Nmap	Identify host availability, port states, service/version clues, and basic OS/traceroute observations.
Nikto	Attempt a web-server configuration and known-issue scan over HTTP.
Burp Suite Community	Use a local proxy listener and inspect HTTP request lines and headers in HTTP history.
curl / getent / ping	Troubleshoot DNS resolution, basic IP reachability, and HTTP/HTTPS connection timeouts.

Method

• Run the assigned Nmap commands against the named training host and save text output under the home directory.

• Attempt a Nikto scan and preserve the connection error rather than interpreting it as a vulnerability result.

• Start Burp Suite's temporary project and verify the local proxy listener at 127.0.0.1:8080.

• Use Burp's built-in browser and HTTP history to view actual request headers.

• Use basic network diagnostics to separate DNS resolution from host/port reachability.

Important limitation

The scan output is a snapshot from this lab session. Filtered ports do not prove that services are absent; they indicate that probes received no response or were filtered. No vulnerability finding should be inferred from a failed connection.

3. Nmap Network Scanning

The assigned target resolved to 44.228.249.3. The host responded to Nmap's host check, but the scan results showed all 1,000 scanned default TCP ports in filtered/no-response states. The reverse DNS name indicated an AWS-hosted endpoint.

3.1 SYN scan

Command: `sudo nmap -sS testphp.vulnweb.com -oN ~/week-nmap-sS.txt`

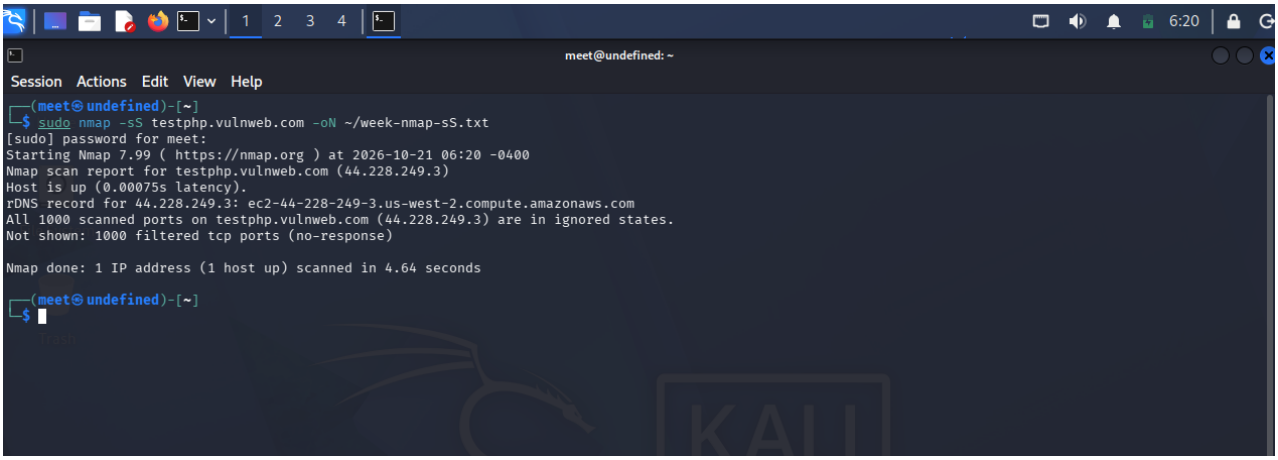


Figure 1. Nmap SYN scan: host up; 1,000 TCP ports filtered.

3.2 Service/version detection

Command: `sudo nmap -sV testphp.vulnweb.com -oN ~/week-nmap-sV.txt`

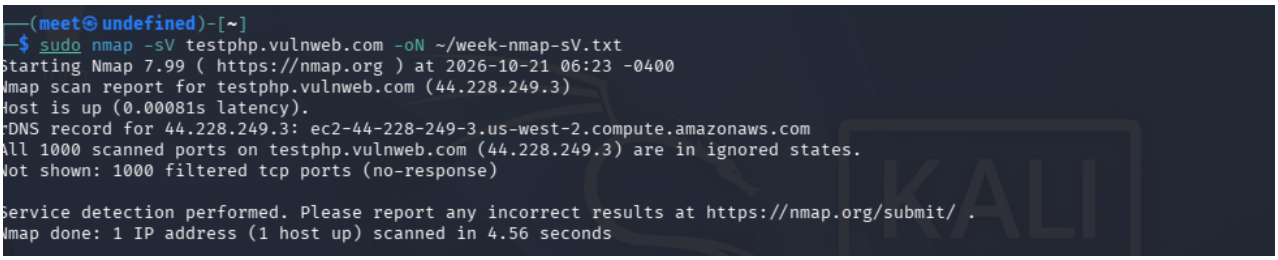


Figure 2. Service/version scan; no open port or service version was identified.

4. Advanced Nmap Scan and Interpretation

Command: `sudo nmap -A testphp.vulnweb.com -oN ~/week-nmap-advanced.txt`

The advanced scan reported the host as up, but all 1,000 scanned TCP ports were filtered. Because Nmap could not find at least one open and one closed port, OS detection was explicitly described as unreliable. It offered several aggressive OS guesses but no exact match; these guesses are not confirmed identification.



Figure 3. Nmap -A output: no exact OS match; OS guesses are unreliable.

What this means

• The target responded sufficiently for Nmap to report the host as up.

• Filtered/no-response results prevent reliable conclusions about the services behind those ports.

• No open TCP ports or confirmed service versions were found in the captured scans.

• The scans do not establish that the target is vulnerable or secure; reachability limited the assessment.

Evidence handling

Text output files were requested by the commands (week-nmap-sS.txt, week-nmap-sV.txt, and week-nmap-advanced.txt). Keep these local output files with the screenshots when assembling the GitHub repository.

5. Nikto Web Vulnerability Scan

Nikto v2.6.0 was launched against the assignment's training URL, with output directed to ~/week5-nikto.txt. The observed result was a connection failure to testphp.vulnweb.com:80. Because the scanner could not establish the HTTP connection, it did not produce usable vulnerability findings in this session.

Observed output

```
[FAIL] Unable to connect to testphp.vulnweb.com:80.
```



Figure 4. Nikto reported that the target's HTTP port could not be reached.

Troubleshooting performed

• Confirmed that the VirtualBox NAT gateway (10.0.2.2) was reachable.

• Confirmed that 1.1.1.1 responded to ping with 0% packet loss.

• Observed inconsistent DNS resolution: the target resolved to 44.228.249.3 in some checks, while example.com lookup timed out in another check.

• HTTP and HTTPS requests to the target timed out after approximately 10 seconds.

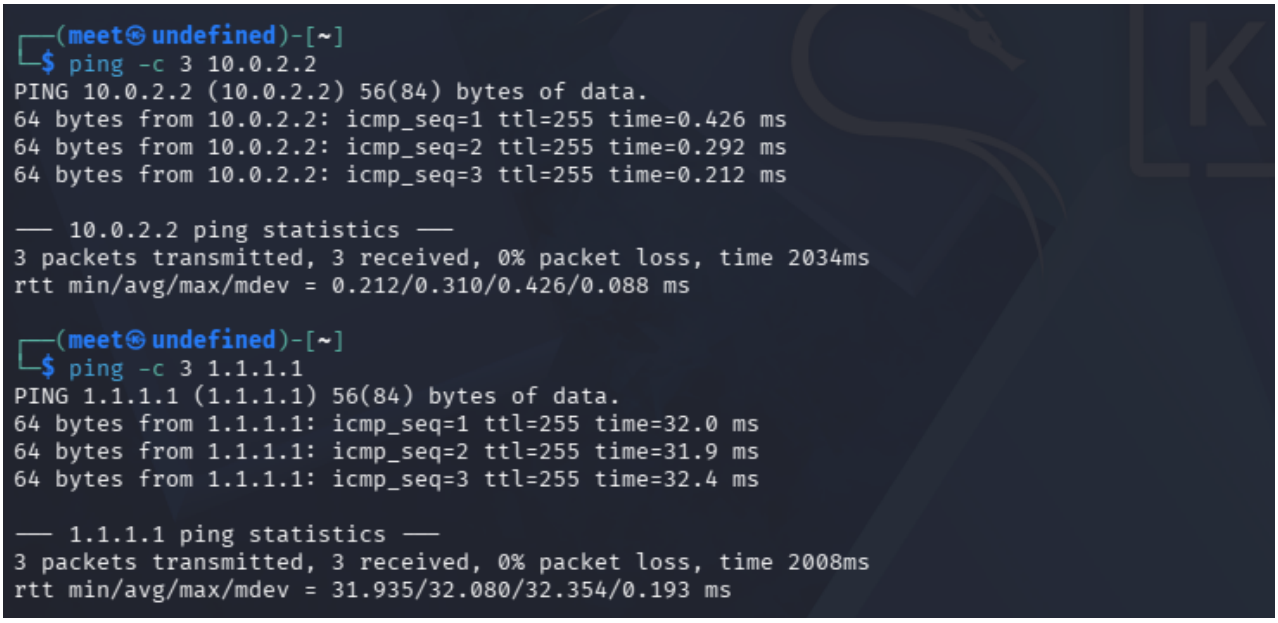


Figure 5. NAT gateway and external IP reachability checks succeeded.

Conclusion for Nikto

This is a connectivity limitation, not a confirmed vulnerability. The correct report outcome is “scan attempted; target connection unavailable; no vulnerability conclusions possible.”

6. Burp Suite Request Inspection

Burp Suite Community Edition was opened using a temporary in-memory project. The proxy listener was enabled on 127.0.0.1:8080. The built-in browser was used to navigate to the target. The target page displayed a Burp-generated connection error, but Burp HTTP history still showed GET requests associated with the target URL.

6.1 Proxy configuration

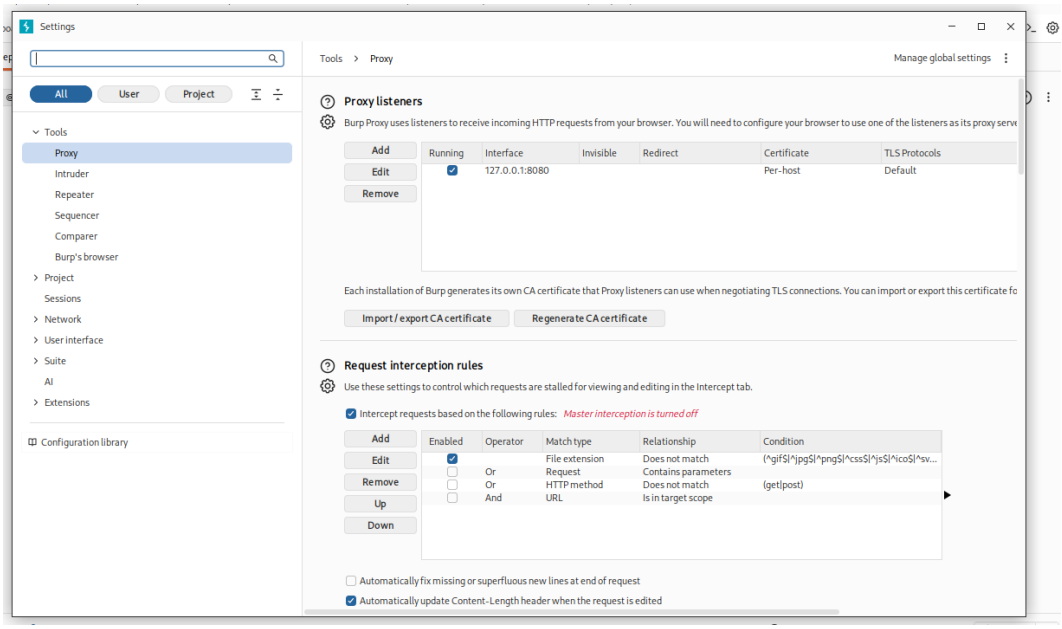


Figure 6. Proxy listener configured at 127.0.0.1:8080.

6.2 Target connection result

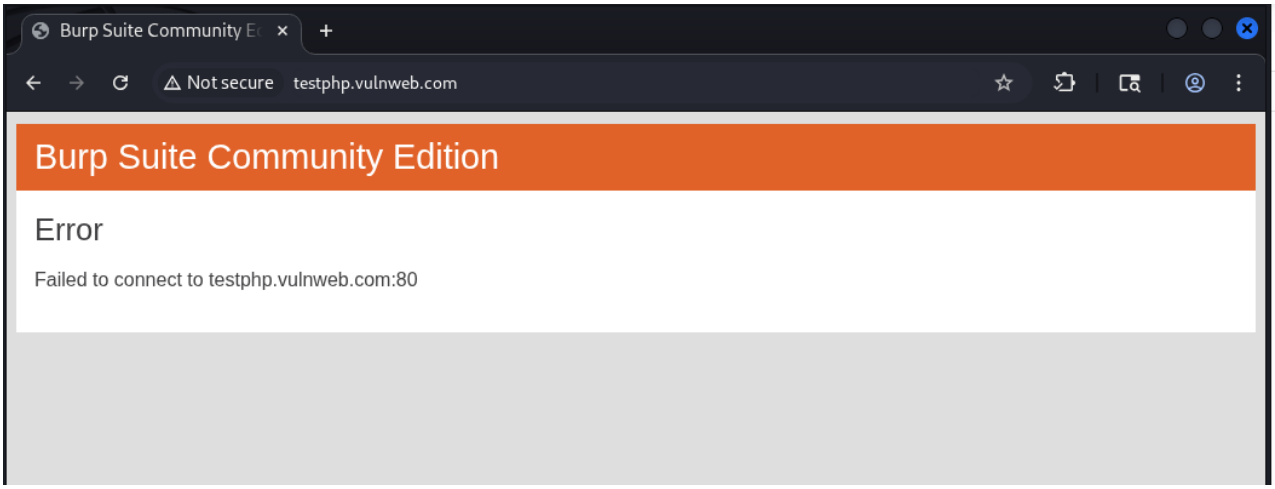


Figure 7. Burp browser displayed a connection error for the target.

7. Burp HTTP History and Request Headers

HTTP history displayed GET / entries for testphp.vulnweb.com. The selected request showed an HTTP/1.1 request line, Host header, language and content-accept headers, a browser User-Agent, and connection keep-alive. These are request-side observations; a successful target response was not captured in the evidence.

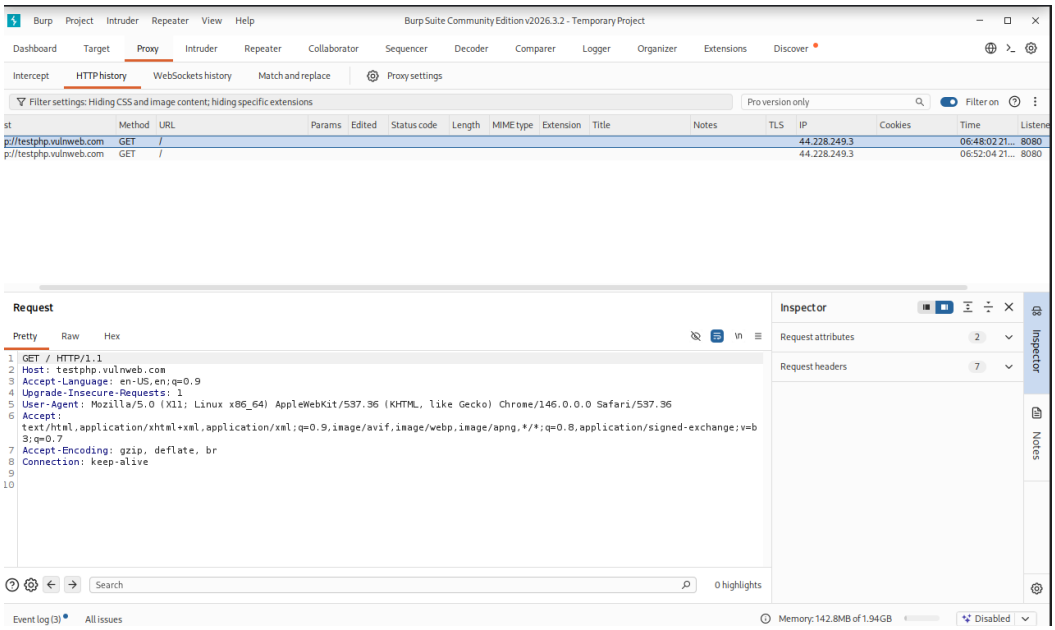


Figure 8. Burp HTTP history shows two GET entries and the selected request headers.

What was learned

• The local Burp listener can receive browser traffic when the browser is configured to use it.

• HTTP history separates requests and allows inspection of method, URL, headers, parameters, and response metadata when available.

• A visible request entry does not prove that the remote server successfully responded.

• Target connection errors should be recorded as limitations, not mislabeled as security findings.

8. Findings, Recommendations, and Conclusion

Findings table

ID	Observation	Severity / status	Evidence-based interpretation
F-01	All 1,000 default TCP ports filtered in captured Nmap scans	Informational	No open ports were confirmed; filtering may limit visibility.
F-02	Advanced OS detection did not find an exact match	Informational	OS guesses were explicitly marked unreliable.
F-03	Nikto could not connect to TCP/80	Limitation	No Nikto vulnerabilities were confirmed.
F-04	HTTP/HTTPS to target timed out	Limitation	Remote web testing was blocked by reachability.
F-05	Burp listener active; GET requests visible in history	Demonstrated / partial	Request inspection demonstrated; successful target response not observed

Recommended next steps

• Retry the assigned training target later or use the course-approved alternate target if the instructor provides one.

• When connectivity is restored, rerun Nikto and save its actual output; do not report unobserved vulnerabilities.

• Capture a successful HTTP response in Burp before describing response codes, cookies, or server headers.

• Keep Nmap text outputs and screenshots together so the results are reproducible.

Conclusion

This practical provided hands-on experience with three common VAPT tools and the importance of validating tool output. Nmap completed but showed filtered ports; Nikto was blocked by target connectivity; Burp's local proxy and request history worked, while the remote target response remained unavailable. The evidence supports a truthful, bounded assessment and highlights how network reachability affects security testing.

End of report